II. $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

We can prove this identity by direct multiplication

We have $(a-b)^3 = (a-b)(a-b)(a-b)$

$= (a-b)^2 \times (a-b)$

$= (a^2 - 2ab + b^2)(a-b)$

$= a(a^2 - 2ab + b^2) - b(a^2 - 2ab + b^2)$

$= a^3 - 2a^2b + ab^2 - ba^2 + 2ab^2 - b^3$

$= a^3 - 2a^2b - ba^2 + ab^2 + 2ab^2 - b^3$

$= a^3 - 3a^2b + 3ab^2 - b^3$

$= RHS$

$(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$

Hence, we proved.

Aliter

$\times$	a^2	$-2ab$	b^2
a	a^3	$-2a^2b$	ab^2
$-b$	$-a^2b$	$2ab^2$	$-b^3$

$= a^3 - 3a^2b + 3ab^2 - b^3$

$a^2b = ba^2$
Multiplication is commutative

III. $(x+a)(x+b)(x+c) = x^3 + (a+b+c)x^2 + (ab+bc+ca)x + abc$

We know that the identity $(x+a)(x+b) = x^2 + (a+b)x + ab$. Let us multiply this by a binomial $(x+c)$. Then we get.

$(x+a)(x+b)(x+c) = [(x+a)(x+b)](x+c)$

$= (x^2 + (a+b)x + ab) \times (x+c)$

$= x[x^2 + (a+b)x + ab] + c[x^2 + (a+b)x + ab]$ (distributive law)

$= x^3 + (a+b)x^2 + abx + cx^2 + (a+b)xc + abc$

$= x^3 + ax^2 + bx^2 + abx + cx^2 + acx + bcx + abc$

$= x^3 + (a+b+c)x^2 + (ab+bc+ca)x + abc$ (Combine x^2, x terms)

$(x+a)(x+b)(x+c) = x^3 + (a+b+c)x^2 + (ab+bc+ca)x + abc$

Thus, we summarise the cubic identities as :

- $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$
- $(a-b)^3 = a^3 - 3a^2b + 3ab^2 - b^3$
- $(x+a)(x+b)(x+c) = x^3 + (a+b+c)x^2 + (ab+bc+ca)x + abc$

Deductions:

The above identities give the following deductions:

(i) $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$

(ii) $a^3 + b^3 = (a+b)(a^2 - ab + b^2)$

(iii) $(a-b)^3 = a^3 - b^3 - 3ab(a-b)$

(iv) $a^3 - b^3 = (a-b)(a^2 + ab + b^2)$

How? Try it!



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